The relatese package

Francisco Mariano Neto

2026/05/30 v0.3

Abstract

relatese is a small helper package for scientific reports. It provides a simplified way of loading commonly used package setups that enhance the standard LaTeX system, with options to load modules for page layout, mathematics, figures, tables, scientific writing, references, and draft tools.

Contents

1 Introduction **1**

2 Installation **2**

3 Basic usage **2**

4 Package options **2**
4.1 The core package . . . 3
4.2 The `layout` option . . . 4
4.3 The `math` option . . . 5
4.3.1 Mathematical commands . . . 5
4.4 The `figures` option . . . 8
4.5 The `tables` option . . . 10
4.6 The `science` option . . . 12
4.7 The `sciencecomma` option . . . 13
4.8 The `refs` option . . . 14
4.9 The `biblatex` option . . . 15
4.10 The `cleveref` option . . . 16
4.11 The `drafttools` option . . . 17
4.12 The `full` option . . . 17

5 Implementation **18**

1 Introduction

relatese is a small helper package for writing scientific reports and related academic documents in LaTeX. It collects a set of defaults, package-loading shortcuts, and notation helpers that are commonly useful in reports containing mathematical expressions, figures, tables, units, chemical formulae, references, and structured page layouts.

The package was originally developed as a personal report template, but it was refactored to be more flexible than a fixed document class or a monolithic preamble. Instead of imposing a complete visual identity or a rigid document structure, relatese provides a compact core and a number of options that trigger the loading of predefined sets of modules. These options can be enabled only when needed, allowing the user to choose, for example, whether to load scientific-writing tools, mathematical notation commands, figure and table helpers, bibliography support, or extended layout features.

The core package loads a conservative set of broadly useful packages and defines a few default conventions for figures, tables, lists, and page geometry. Additional functionality is activated through package options such as `math`, `figures`, `tables`, `science` (with `sciencecomma` as an alternative), `refs`, and `full`. This makes the package suitable both for short reports with minimal setup and for larger scientific documents requiring more infrastructure.

One design goal of relatese is to keep common report-writing choices close at hand while avoiding the need to copy the same long preamble from project to project. Another goal is to keep these choices explicit: features that can affect compilation requirements or document semantics, such as svg, biblatex, mhchem, or cleveref, are grouped behind options rather than being treated as unavoidable defaults.

Although the package includes conveniences that are especially useful in scientific and technical writing, it is not meant to replace specialized document classes, journal templates, or institutional formatting rules. When a publisher, university, or funding agency provides its own class or template, that template should take precedence. In such cases, relatese can still be useful as a lightweight collection of helpers, provided that the selected options do not conflict with the required formatting.

The package also includes support for documents following Portuguese numerical conventions through the `sciencecomma` option, which configures siunitx to use a decimal comma. Language-dependent behavior is otherwise kept limited, so that document language can be handled by standard tools such as babel or polyglossia.

In short, relatese is intended to be a practical middle ground between a bare LaTeX preamble and a fully custom document class, providing provides reusable defaults and helpers, while leaving the document class, language setup, and final visual identity under the author's control.

2 Installation

Run LaTeX on `relatese.ins` to generate `relatese.sty`:

```
latex relatese.ins
```

Then place `relatese.sty` where TeX can find it, or keep it in the same directory as your document.

This documentation can be generated by running:

```
latex relatese.dtx
```

3 Basic usage

The package can be loaded without options:

```
\usepackage{relatese}
```

A fuller setup for scientific reports might use:

```
\usepackage[layout,math,figures,tables,sciencecomma,refs]{relatese}
```

4 Package options

relatese is organized around a small core and a set of optional modules. The core is always loaded and provides general-purpose defaults for page geometry, graphics, captions, tables, lists, colors, and basic mathematical support through amsmath. Additional features are enabled through package options.

Option	Effect
`layout`	Loads document-structure helpers.
`math`	Loads extra mathematical packages and notation helpers.
`figures`	Loads float, subfigure, landscape, barrier, and SVG support.
`tables`	Loads colored-table and cell-formatting helpers.
`science`	Loads `siunitx` and `mhchem`.
`sciencecomma`	Like `science`, with decimal comma for `siunitx`.
`siunitx`	Loads `siunitx` only.
`chemistry`	Loads `mhchem` only.
`refs`	Loads `biblatex`, `hyperref`, and `cleveref`.
`biblatex`	Loads `csquotes` and `biblatex`.
`cleveref`	Loads `hyperref` and `cleveref`.
`drafttools`	Loads drafting utilities such as `lipsum`.
`full`	Enables the main optional modules.

Options may be combined in the usual LaTeX way:

```
\usepackage[math,sciencecomma,refs]{relatese}
```

For convenience, the `full` option enables most modules at once. This is useful for larger reports or for documents following the conventions for which the package was originally designed. For more portable documents, however, it is usually better to enable only the modules that are actually needed.

4.1 The core package

Loading relatese without options activates the core configuration:

```
\usepackage{relatese}
```

The core loads a conservative set of packages that are broadly useful in scientific reports and academic writing: xcolor, microtype, graphicx, amsmath, bm, booktabs, caption, enumitem, and geometry.

The core also defines a small set of colors, sets a default graphics search path, configures captions for figures and tables, applies compact spacing to `itemize` lists, and makes page geometry presets available through `\setgeometry`. Since geometry is always loaded, these presets are available even when no package option is specified.

Example:

```
\documentclass[a4paper]{article}
\usepackage{relatese}

\begin{document}

\section{Introduction}

This document uses the core defaults provided by \pkg{relatese}.
```

```
\begin{itemize}
  \item Compact item spacing is already configured.
  \item Figure and table captions use the package defaults.
  \item Geometry presets are available through \cs{setgeometry}.
\end{itemize}

\end{document}
```

The default geometry preset is `standard`. It changes only the margins and does not force a paper size. The paper size should be selected by the document class, for example:

```
\documentclass[a4paper]{article}
\usepackage{relatese}
```

or:

```
\documentclass[letterpaper]{article}
\usepackage{relatese}
```

4.2 The `layout` option

The `layout` option enables additional packages related to document structure and page layout:

```
\usepackage[layout]{relatese}
```

It loads changepage, multicol, authblk, setspace, indentfirst, tocloft, lastpage, titlesec, and footmisc.

This option is intended for reports whose overall layout is controlled by the author. It is useful when the document needs multiple columns, customized author blocks, line spacing commands, first-paragraph indentation, table-of-contents adjustments, page-count references, or section-title formatting.

Since `layout` changes document-wide conventions, it should be used with some care when working with publisher templates, institutional classes, or document classes that already customize section headings, footnotes, or the table of contents.

In particular, `layout` changes footnote numbering to symbolic marks:

```
*, dagger, double dagger, ...
```

This reproduces one of the original conventions of the package, but it may not be appropriate for all document types.

Example:

```
\documentclass[a4paper]{article}
\usepackage[layout]{relatese}

\title{A Scientific Report}
\author[1]{First Author}
\author[2]{Second Author}
\affil[1]{Institute A}
\affil[2]{Institute B}

\begin{document}
```

```
\maketitle
\onehalfspacing

\begin{multicols}{2}
This section is typeset in two columns.
\end{multicols}

\end{document}
```

The package includes a few geometry presets, summarized in table 4.1. None of them sets the paper size. Paper size should be selected by the document class, for example with `a4paper` or `letterpaper`.

Table 4.1: Geometry presets provided by relatese.

Preset	Margins	Intended use
`standard`	`hmargin=2.5cm, vmargin=3.5cm`	General reports and default use.
`compact`	`hmargin=2cm, vmargin=2.5cm`	Shorter documents or drafts with less white space.
`wide`	`hmargin=2cm, top=2.5cm, bottom=3cm`	Documents needing a wider text block, for example when equations, figures, or tables benefit from additional horizontal space.
`binding`	`inner=3cm, outer=2cm, top=3cm, bottom=3.5cm`	Printed documents that require extra inner margin for binding.
`draft`	`hmargin=3cm, vmargin=3.5cm`	Drafts with larger margins for handwritten notes, comments, or corrections.

These presets may be selected in the preamble with `\setgeometry`. For temporary geometry changes inside the document body, use `\newsetgeometry` and return to the previous layout with `\restoregeometry`.

4.3 The `math` option

The `math` option enables additional mathematical packages and a collection of notation helpers:

```
\usepackage[math]{relatese}
```

It loads amssymb, xfrac, cancel, and xparse. It also defines commands for common mathematical notation, including derivatives, differentials, inner products, chi-square symbols, Roman numerals, prime marks, and unit vectors.

4.3.1 Mathematical commands

The `math` option defines a small set of commands for common mathematical notation. The commands are intended as lightweight shortcuts for notation frequently used in scientific reports, lecture notes, and technical documents.

Basic notation

Command	Example input	Output
`\mean`	`\mean{x}`	$\langle x \rangle$
`\lapla`	`\lapla f`	$\nabla^2 f$
`\relprime`	`x\relprime`	x'
`\relprime` with argument	`x\relprime[2]`	x'^2
`\slparallel`	`E_{\slparallel}`	$E_{/\!/}$

The command `\relprime` is provided instead of a more generic name such as `\prime`, since `\prime` is already a standard mathematical symbol in LaTeX. With an optional argument, `\relprime` can be used to indicate repeated primes.

The command `\slparallel` provides a slanted parallel marker. It does not redefine the standard `\parallel` symbol, so both forms remain available.

Differentials and derivatives

Command	Example input	Output
`\ud`	`\ud x`	$\mathrm{d}x$
`\ud` with order	`\ud[2]x`	d^2x
`\dd`	`\dd{f}{x}`	$\frac{\mathrm{d}f}{\mathrm{d}x}$
`\dd` with order	`\dd{f}{x}[2]`	$\frac{\mathrm{d}^2f}{\mathrm{d}x^2}$
`\dda`	`\dda{f}{x}`	$\frac{\mathrm{d}}{\mathrm{d}x}f$
`\dda` with order	`\dda{f}{x}[2]`	$\frac{\mathrm{d}^2}{\mathrm{d}x^2}f$
`\ddp`	`\ddp{f}{x}`	$\frac{\partial f}{\partial x}$
`\ddp` with order	`\ddp{f}{x}[2]`	$\frac{\partial^2 f}{\partial x^2}$
`\ddpa`	`\ddpa{f}{x}`	$\frac{\partial}{\partial x}f$
`\ddpa` with order	`\ddpa{f}{x}[2]`	$\frac{\partial^2}{\partial x^2}f$

The commands `\dd` and `\ddp` typeset total and partial derivatives of a given expression. The commands `\dda` and `\ddpa` typeset the corresponding differential operators applied to an expression. The optional argument specifies the order of the derivative.

For example:

```
\[
  \dd{f}{x}
  \qquad
  \dd{f}{x}[2]
  \qquad
  \ddp{f}{t}
\]
```

produces:

$$\frac{\mathrm{d}f}{\mathrm{d}x} \qquad \frac{\mathrm{d}^2f}{\mathrm{d}x^2} \qquad \frac{\partial f}{\partial t}$$

Products, statistics, and numerals

Command	Example input	Output
`\innp`	`\innp{u}{v}`	$\langle u, v \rangle$
`\chisq`	`\chisq`	χ^2
`\chisqred`	`\chisqred`	χ^2_{red}
`\rmnum`	`\rmnum{4}`	iv
`\Rmnum`	`\Rmnum{4}`	IV

The command `\innp` typesets a simple inner product using angle brackets. The commands `\chisq` and `\chisqred` are shortcuts for chi-square and reduced chi-square notation. The commands `\rmnum` and `\Rmnum` produce lower-case and upper-case Roman numerals.

Unit vectors

The package also defines shortcuts for common Cartesian, cylindrical, spherical, perpendicular, and parallel unit vectors.

Command	Example input	Output
`\vi`	`\vi`	$\hat{e}_1$
`\vj`	`\vj`	$\hat{e}_2$
`\vk`	`\vk`	$\hat{e}_3$
`\vs`	`\vs`	$\hat{e}_s$
`\vp`	`\vp`	$\hat{e}_p$
`\vperp`	`\vperp`	$\hat{e}_\perp$
`\vpar`	`\vpar`	$\hat{e}_{/\!/}$
`\vphi`	`\vphi`	$\hat{e}_\phi$
`\vz`	`\vz`	$\hat{e}_z$
`\vsl`	`\vsl`	$\hat{e}_{s'}$
`\vphil`	`\vphil`	$\hat{e}_{\phi'}$
`\vzl`	`\vzl`	$\hat{e}_{z'}$
`\vr`	`\vr`	$\hat{e}_r$
`\vtheta`	`\vtheta`	$\hat{e}_\theta$

For example:

```
\[
  \vec{A}
  =
  A_{\perp}\vperp
  +
  A_{\slparallel}\vpar
\]
```

produces:

$$\vec{A} = A_{\perp}\hat{e}_{\perp} + A_{/\!/}\hat{e}_{/\!/}$$

Display-math spacing

The command `\seteqskip` adjusts the vertical spacing around displayed equations:

```
\seteqskip
```

It sets `\abovedisplayskip`, `\belowdisplayskip`, `\abovedisplayshortskip`, and `\belowdisplayshortskip`. It is intended to be used after choosing the document line spacing, for example after `\onehalfspacing` or `\doublespacing`.

4.4 The `figures` option

The `figures` option enables additional support for figures, subfigures, float placement, landscape pages, and SVG graphics:

```
\usepackage[figures]{relatese}
```

This option loads float, subcaption, placeins, pdflscape, and svg. It is intended for documents that make substantial use of figures or that need more control over figure placement than the LaTeX defaults provide.

The core package already loads graphicx and caption, so basic figure inclusion is available even without this option. The `figures` option adds tools for subfigures, float barriers, landscape pages, and SVG inclusion.

Basic figure inclusion. Since graphicx is part of the core package, ordinary figures may be included without enabling `figures`:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=.65\linewidth]{example-image}
  \caption{A simple figure.}
  \label{fig:simple}
\end{figure}
```

This produces a standard floating figure with the caption style configured by relatese. The package also adds `img/` to the graphics search path, so image files placed in that directory can be included without writing the full path.

Subfigures. With `figures`, the subcaption package is available. This allows several related panels to be grouped inside a single figure:

```
\begin{figure}[htbp]
  \centering

  \begin{subfigure}{.45\linewidth}
    \centering
    \includegraphics[width=\linewidth]{example-image-a}
    \caption{First panel.}
    \label{fig:panel-a}
  \end{subfigure}
```

```
  \hfill
  \begin{subfigure}{.45\linewidth}
    \centering
    \includegraphics[width=\linewidth]{example-image-b}
    \caption{Second panel.}
    \label{fig:panel-b}
  \end{subfigure}

  \caption{A figure composed of two panels.}
  \label{fig:two-panels}
\end{figure}
```

Subfigure captions are configured to use the same general caption style as ordinary figures.

Float barriers. The placeins package provides the command `\FloatBarrier`. It prevents floats from moving past a given point in the document:

```
\section{Results}

Text discussing the first group of figures.

\begin{figure}[htbp]
  \centering
  \includegraphics[width=.7\linewidth]{result}
  \caption{A result figure.}
\end{figure}

\FloatBarrier

\section{Discussion}
```

This is useful in reports where figures should remain close to the section in which they are discussed.

Forced float placement. The float package provides the `H` placement specifier:

```
\begin{figure}[H]
  \centering
  \includegraphics[width=.7\linewidth]{setup}
  \caption{A figure placed exactly here.}
\end{figure}
```

This can be useful in drafts, laboratory reports, or teaching material. However, forced placement should be used sparingly, since it can make page breaking more difficult and may produce poor page layouts.

Landscape pages. The pdflscape package is useful for wide figures, diagrams, or tables that do not fit comfortably in portrait orientation:

```
\begin{landscape}
\begin{figure}[htbp]
  \centering
  \includegraphics[width=.9\linewidth]{wide-diagram}
  \caption{A wide diagram on a landscape page.}
\end{figure}
\end{landscape}
```

Pages inside the `landscape` environment are rotated in the output PDF, which is convenient for on-screen reading.

SVG graphics. The `figures` option also loads `svg`, allowing SVG files to be included with `\includesvg`:

```
\begin{figure}[htbp]
  \centering
  \includesvg[width=.65\linewidth]{diagram}
  \caption{A figure included from an SVG file.}
\end{figure}
```

In many workflows, SVG inclusion requires external conversion tools such as Inkscape, and may require shell escape depending on the compilation setup. This should be checked when building documents in restricted or automated environments.

Example. The following minimal example uses subfigures and a float barrier:

```
\documentclass[a4paper]{article}
\usepackage[figures]{relatese}

\begin{document}

\section{Experimental setup}

\begin{figure}[htbp]
  \centering

  \begin{subfigure}{.45\linewidth}
    \centering
    \includegraphics[width=\linewidth]{example-image-a}
    \caption{First view.}
  \end{subfigure}
  \hfill
  \begin{subfigure}{.45\linewidth}
    \centering
    \includegraphics[width=\linewidth]{example-image-b}
    \caption{Second view.}
  \end{subfigure}

  \caption{Two views of the experimental setup.}
\end{figure}

\FloatBarrier

\section{Results}

The figures from the previous section will not float into this section.

\end{document}
```

4.5 The `tables` option

The `tables` option enables additional tools for more elaborate tables:

```
\usepackage[tables]{relatese}
```

This option loads colortbl and makecell. The core package already loads booktabs, so high-quality horizontal rules are available even without `tables`. The `tables` option is intended for documents that need colored rows or cells, multi-line table cells, or more flexible table headings.

Basic tables. Since booktabs is part of the core package, simple tables can be written without enabling `tables`:

```
\begin{table}[htbp]
\centering
\caption{A simple table using \pkg{booktabs}.}
\begin{tabular}{ll}
\toprule
Quantity & Value \\
\midrule
A & 1.0 \\
B & 2.0 \\
\bottomrule
\end{tabular}
\end{table}
```

The `tables` option adds tools for cases where this basic structure is not enough.

Multi-line cells. The makecell package provides the command `\makecell`, which is useful for table headings or cells that need explicit line breaks:

```
\begin{table}[htbp]
\centering
\caption{A table using multi-line cells.}
\begin{tabular}{ll}
\toprule
Quantity & \makecell{Measured\\value} \\
\midrule
A & 1.0 \\
B & 2.0 \\
\bottomrule
\end{tabular}
\end{table}
```

Colored rows and cells. The colortbl package allows rows and cells to be colored. The package colors defined by relatese, such as `superlightgray`, may be used for subtle table backgrounds:

```
\begin{table}[htbp]
\centering
\caption{A table with a highlighted row.}
\begin{tabular}{ll}
\toprule
Quantity & Value \\
\midrule
\rowcolor{superlightgray}
A & 1.0 \\
B & 2.0 \\
\bottomrule
\end{tabular}
\end{table}
```

Table coloring should be used with care. In particular, colors should not be the only way to communicate information, since the document may be printed in grayscale or read with accessibility tools.

Example. The following example combines booktabs, makecell, and colortbl:

```
\documentclass[a4paper]{article}
\usepackage[tables]{relatese}

\begin{document}

\begin{table}[htbp]
\centering
\caption{Example table using the \opt{tables} option.}
\begin{tabular}{lll}
\toprule
Sample & \makecell{Measured\\quantity} & Value \\
\midrule
\rowcolor{superlightgray}
A & Thickness & 10.0 \\
B & Thickness & 12.5 \\
C & Thickness & 11.8 \\
\bottomrule
\end{tabular}
\end{table}

\end{document}
```

4.6 The `science` option

The `science` option enables tools commonly used in scientific and technical writing:

```
\usepackage[science]{relatese}
```

This option loads siunitx and mhchem. It is intended for documents that contain physical quantities, units, chemical formulae, or chemical reactions.

Units and quantities. The siunitx package provides consistent formatting for numbers, units, ranges, and quantities:

```
The sample thickness was \SI{10}{\micro\meter}.
The beam energy was \SI{2.4}{\mega\electronvolt}.
The speed was \SI{3.0e8}{\meter\per\second}.
```

The `science` option applies a small set of defaults:

```
group-digits=none,
range-phrase=--,
per-mode=symbol
```

This means that ranges and units can be written as:

```
\SIrange{1}{5}{\mega\electronvolt}
\si{\meter\per\second}
```

Chemical notation. The mhchem package is loaded with version 4 syntax. It provides the command `\ce` for chemical formulae, ions, isotopes, and reactions:

```
\ce{H2O}
\ce{Fe^{2+}}
\ce{^{60}Co}
\ce{A + B -> C}
```

For example:

```
Water radiolysis may be represented schematically as
\[
  \ce{H2O -> H^. + OH^.}
\]
```

Example. The following example uses both siunitx and mhchem:

```
\documentclass[a4paper]{article}
\usepackage[science]{relatese}

\begin{document}

The target thickness was \SI{10}{\micro\meter}, and the proton
beam energy was \SI{2.4}{\mega\electronvolt}.

A simple reaction may be written as
\[
  \ce{H2O -> H^. + OH^.}
\]

\end{document}
```

The `science` option is useful for reports in physics, chemistry, engineering, medical physics, and related fields. Documents that follow decimal-comma conventions should use `sciencecomma` instead.

4.7 The `sciencecomma` option

The `sciencecomma` option is a variant of `science` for documents that use a decimal comma:

```
\usepackage[sciencecomma]{relatese}
```

It loads the same scientific-writing tools as `science`, but also configures siunitx to use a comma as the decimal marker:

```
\sisetup{output-decimal-marker={,}}
```

This is useful for documents written according to Portuguese or other decimal-comma conventions:

```
\SI{3.14}{\meter}
```

will be printed using a decimal comma according to the siunitx configuration.

Language selection itself is not handled by relatese. Users should continue to use babel or polyglossia for hyphenation, translated strings, and other language-dependent behavior.

4.8 The `refs` option

The `refs` option enables the complete reference-management setup provided by relatese:

```
\usepackage[refs]{relatese}
```

This option loads csquotes, biblatex, hyperref, and cleveref. It is intended for documents whose bibliography, hyperlinks, and cross-reference style are controlled by the author.

Bibliography setup. The bibliography is handled by biblatex with `biber` as the backend. The default options are:

```
backend=biber,
style=numeric-comp,
sorting=none,
backref=true,
autocite=superscript
```

This gives numeric compressed citations, bibliography entries sorted by order of appearance, backreferences, and superscript autocitations.

A typical document uses:

```
\addbibresource{references.bib}
...
\autocite{key}
...
\printbibliography
```

Documents using this option should be compiled with `biber`, not BibTEX.

Hyperlinks. The option also loads hyperref with Unicode bookmarks, breakable links, and colored links:

```
colorlinks=true,
linkcolor=darkblue,
urlcolor=darkblue,
citecolor=darkblue
```

The color `darkblue` is defined by the core package.

Clever references. The cleveref package is loaded after hyperref. This allows references such as:

```
\cref{sec:method}
\Cref{fig:setup}
```

No explicit language option is passed to cleveref. The document language should be configured with babel or polyglossia.

Example.

```
\documentclass[a4paper]{article}
\usepackage[refs]{relatese}

\addbibresource{references.bib}

\begin{document}

\section{Introduction}

As discussed in \autocite{einstein1905}, references are handled by
\pkg{biblatex}.

The method is described in \cref{sec:method}.

\section{Method}
\label{sec:method}

\printbibliography

\end{document}
```

Since reference and bibliography styles are often specified by journals, publishers, or universities, the `refs` option should be used only when these choices are under the author's control.

4.9 The `biblatex` option

The `biblatex` option enables only the bibliography part of the reference setup:

```
\usepackage[biblatex]{relatese}
```

This option loads csquotes and biblatex, using the same bibliography configuration as the `refs` option:

```
backend=biber,
style=numeric-comp,
sorting=none,
backref=true,
autocite=superscript
```

It does not load hyperref or cleveref. This is useful when the document should use the relatese bibliography defaults, but hyperlinks or clever references need to be configured separately.

Example.

```
\documentclass[a4paper]{article}
\usepackage[biblatex]{relatese}

\addbibresource{references.bib}

\begin{document}

A citation appears here \autocite{key}.

\printbibliography

\end{document}
```

Documents using this option should be compiled with `biber`. If hyperlinks and intelligent cross-references are also desired, use `refs` instead.

4.10 The `cleveref` option

The `cleveref` option enables intelligent cross-references:

```
\usepackage[cleveref]{relatese}
```

It loads cleveref. If hyperref has not already been loaded through `refs`, relatese loads hyperref first. This preserves the usual loading order:

```
hyperref
cleveref
```

The option is useful for documents with many labeled sections, figures, tables, or equations.

Basic usage. With cleveref, references can be written without manually typing words such as "section", "figure", or "table":

```
See \cref{sec:method}.
See \Cref{fig:setup}.
```

The package automatically determines the type of the referenced object.

Language. No explicit language option is passed to cleveref. Language selection should be handled by babel or polyglossia. This keeps relatese from forcing a specific document language.

Example.

```
\documentclass[a4paper]{article}
\usepackage[cleveref]{relatese}

\begin{document}

\section{Experimental setup}
\label{sec:setup}

The experimental setup is described in \cref{sec:setup}.

\begin{figure}[htbp]
  \centering
  \rule{.5\linewidth}{2cm}
  \caption{A placeholder figure.}
  \label{fig:placeholder}
\end{figure}

See also \Cref{fig:placeholder}.

\end{document}
```

If bibliography management is also desired, use `refs` instead of enabling `cleveref` alone.

4.11 The `drafttools` option

The `drafttools` option loads utilities that are useful while drafting or testing a document layout:

```
\usepackage[drafttools]{relatese}
```

Currently, this option loads lipsum, which provides dummy text. Dummy text is useful when testing page layout, headings, figures, tables, and section structure before the final content is available.

Example.

```
\documentclass[a4paper]{article}
\usepackage[drafttools]{relatese}

\begin{document}

\section{Draft section}

\lipsum[1-3]

\end{document}
```

Since dummy-text tools are not needed in final documents, they are kept behind an explicit option rather than being loaded by default.

4.12 The `full` option

The `full` option enables the main working set of modules provided by relatese:

```
\usepackage[full]{relatese}
```

It is intended for documents that use most of the package conventions: layout helpers, mathematical notation, figure and SVG support, extended tables, scientific-writing tools, bibliography support, hyperlinks, and clever references.

In practice, `full` is equivalent to enabling the main options:

```
layout,
math,
figures,
tables,
sciencecomma,
refs
```

The option is convenient for personal reports, teaching material, or scientific documents whose formatting is controlled by the author. For documents subject to journal, publisher, or institutional requirements, it is usually better to enable only the specific options needed.

Example.

```
\documentclass[a4paper]{article}
\usepackage[full]{relatese}

\addbibresource{references.bib}

\begin{document}
```

```
\section{Introduction}

A quantity with units may be written as
\SI{10}{\micro\meter}.

A derivative may be written as
\[
  \dd{f}{x}
\]

The method is described in \cref{sec:method}.

\section{Method}
\label{sec:method}

A chemical expression may be written as
\[
  \ce{H2O -> H^. + OH^.}
\]

\printbibliography

\end{document}
```

Documents using `full` and bibliography commands should be compiled with `biber`.

5 Implementation

```
%% relatese.sty -- General-purpose report helper package
%% Reorganized version with option-based package loading.
%%
%% Suggested usage:
%%   \usepackage{relatese}
%%   \usepackage[layout,sciencecomma,refs]{relatese}
%%   \usepackage[full]{relatese}
%%
%% Geometry presets:
%%   \setgeometry{standard}
%%   \setgeometry{compact}
%%   \setgeometry{wide}
%%   \setgeometry{binding}
%%   \setgeometry{draft}

\NeedsTeXFormat{LaTeX2e}
\ProvidesPackage{relatese}[2026/05/30 v0.3 General report helper package]

\makeatletter

% ------------------------------------------------------------------------
% Option switches
% ------------------------------------------------------------------------

\newif\ifrelatese@layout
\newif\ifrelatese@math
\newif\ifrelatese@figures
\newif\ifrelatese@tables
```

```
\newif\ifrelatese@siunitx
\newif\ifrelatese@sciencecomma
\newif\ifrelatese@chemistry
\newif\ifrelatese@refs
\newif\ifrelatese@biblatex
\newif\ifrelatese@cleveref
\newif\ifrelatese@drafttools

% Core packages are always loaded. They are small, broadly useful, and unlikely
% to impose a document structure.

% Layout and document-structure options.
\DeclareOption{layout}{%
  \relatese@layouttrue
}

% Mathematics options.
\DeclareOption{math}{\relatese@mathtrue}

% Figure and table options.
\DeclareOption{figures}{%
  \relatese@figurestrue
}
\DeclareOption{tables}{\relatese@tablestrue}

% Scientific-writing options. The science option loads siunitx and mhchem;
% sciencecomma also enables decimal comma in siunitx.
\DeclareOption{science}{%
  \relatese@siunitxtrue
  \relatese@chemistrytrue
}
\DeclareOption{sciencecomma}{%
  \relatese@siunitxtrue
  \relatese@chemistrytrue
  \relatese@sciencecommatrue
}
\DeclareOption{siunitx}{\relatese@siunitxtrue}
\DeclareOption{chemistry}{\relatese@chemistrytrue}

% Reference-management options.
\DeclareOption{refs}{%
  \relatese@refstrue
  \relatese@biblatextrue
  \relatese@clevereftrue
}
\DeclareOption{biblatex}{\relatese@biblatextrue}
\DeclareOption{cleveref}{\relatese@clevereftrue}

% Development option.
\DeclareOption{drafttools}{\relatese@drafttoolstrue}

% Convenience option: approximates the older monolithic behavior.
\DeclareOption{full}{%
  \relatese@layouttrue
  \relatese@mathtrue
  \relatese@figurestrue
  \relatese@tablestrue
  \relatese@siunitxtrue
```

```
  \relatese@chemistrytrue
  \relatese@refstrue
  \relatese@biblatextrue
  \relatese@clevereftrue
}

\DeclareOption*{%
  \PackageWarning{relatese}{Unknown option `\CurrentOption'}%
}

\ProcessOptions\relax

% ----------------------------------------------------------------------------
% Core package set
% ----------------------------------------------------------------------------

\RequirePackage{xcolor}
\RequirePackage{microtype}
\RequirePackage{graphicx}
\RequirePackage{amsmath}
\RequirePackage{bm}
\RequirePackage{booktabs}
\RequirePackage[hang,footnotesize,bf,tableposition=top]{caption}
\RequirePackage{enumitem}

% Additional colors to what is defined in xcolor
\definecolor{darkblue}{RGB}{0,0,192}
\definecolor{lightblue}{RGB}{48,224,255}
\definecolor{darkgreen}{RGB}{0,128,0}
\definecolor{darkred}{RGB}{192,0,0}
\definecolor{lightred}{RGB}{224,92,92}
\definecolor{orange}{RGB}{254,80,0}
\definecolor{wine}{RGB}{142,54,70}
\definecolor{superlightgray}{RGB}{240,240,240}
\definecolor{darkergray}{RGB}{80,80,80}

% Default graphic search path. This is convenient, but still mild enough to keep
% in the core because it only adds a lookup path.
\graphicspath{{img/}}
\DeclareGraphicsExtensions{.pdf,.png,.jpg,.jpeg,.eps}

% Caption defaults.
\captionsetup[figure]{font=footnotesize,format=hang}
\captionsetup[table]{font=footnotesize,format=hang,position=above}

% Compact itemize lists.
\setlist[itemize]{%
  itemsep=2pt,
  topsep=10pt,
  parsep=0pt,
  partopsep=0pt
}

% ----------------------------------------------------------------------------
% Layout and page geometry
% ----------------------------------------------------------------------------

\RequirePackage{geometry}
```

```
145
146
147 % Select a named geometry preset. This command is intended for the preamble.
148 \newcommand{\setgeometry}[1]{%
149   \@ifundefined{relatese@geometry@#1}{%
150     \PackageError{relatese}
151     {Unknown geometry preset: #1}
152     {Available presets: standard, compact, wide, binding, draft.}%
153   }{%
154     \csname relatese@geometry@#1\endcsname
155   }%
156 }
157
158 % Apply a named geometry preset in the document body, using geometry's
159 % \newgeometry. Return to the previous layout with \restoregeometry.
160 \newcommand{\newsetgeometry}[1]{%
161   \@ifundefined{relatese@newgeometry@#1}{%
162     \PackageError{relatese}
163     {Unknown geometry preset: #1}
164     {Available presets: standard, compact, wide, binding, draft.}%
165   }{%
166     \csname relatese@newgeometry@#1\endcsname
167   }%
168 }
169
170 \newcommand{\relatese@geometry@standard}{%
171   \geometry{hmargin=2.5cm,vmargin=3.5cm}%
172 }
173 \newcommand{\relatese@geometry@compact}{%
174   \geometry{hmargin=2cm,vmargin=2.5cm}%
175 }
176 \newcommand{\relatese@geometry@wide}{%
177   \geometry{hmargin=2cm,top=2.5cm,bottom=3cm}%
178 }
179 \newcommand{\relatese@geometry@binding}{%
180   \geometry{inner=3cm,outer=2cm,top=3cm,bottom=3.5cm}%
181 }
182 \newcommand{\relatese@geometry@draft}{%
183   \geometry{hmargin=3cm,vmargin=3.5cm}%
184 }
185
186 \newcommand{\relatese@newgeometry@standard}{%
187   \newgeometry{hmargin=2.5cm,vmargin=3.5cm}%
188 }
189 \newcommand{\relatese@newgeometry@compact}{%
190   \newgeometry{hmargin=2cm,vmargin=2.5cm}%
191 }
192 \newcommand{\relatese@newgeometry@wide}{%
193   \newgeometry{hmargin=2cm,top=2.5cm,bottom=3cm}%
194 }
195 \newcommand{\relatese@newgeometry@binding}{%
196   \newgeometry{inner=3cm,outer=2cm,top=3cm,bottom=3.5cm}%
197 }
198 \newcommand{\relatese@newgeometry@draft}{%
199   \newgeometry{hmargin=3cm,vmargin=3.5cm}%
200 }
201
202 % Apply the default geometry preset.
```

```
\setgeometry{standard}

\ifrelatese@layout
  \RequirePackage{changepage}
  \RequirePackage{multicol}
  \RequirePackage{authblk}
  \RequirePackage{setspace}
  \RequirePackage{indentfirst}
  \RequirePackage{tocloft}
  \RequirePackage{lastpage}
  \RequirePackage{titlesec}
  \RequirePackage{footmisc}

  % Run-in unnumbered paragraphs.
  \titleformat{name=\paragraph,numberless}[runin]%
    {\normalfont\normalsize\bfseries}{}{5pt}{}

  % Symbol footnotes reproduce the original package behavior. This is kept under
  % the layout option because it changes document-wide numbering.
  \renewcommand{\thefootnote}{\fnsymbol{footnote}}
\fi

% ------------------------------------------------------------------------------
% Mathematics
% ------------------------------------------------------------------------------

\ifrelatese@math
  \RequirePackage{amssymb}
  \RequirePackage{xfrac}
  \RequirePackage{cancel}
  \renewcommand{\CancelColor}{\color{red}}

  \RequirePackage{xparse}

  % Display math spacing helper. Use after setting line spacing.
  \newcommand{\seteqskip}{%
    \setlength{\abovedisplayskip}{15pt}%
    \setlength{\belowdisplayskip}{25pt}%
    \setlength{\abovedisplayshortskip}{10pt}%
    \setlength{\belowdisplayshortskip}{20pt}%
  }

  % Common notation helpers.
  \newcommand{\mean}[1]{\ensuremath{\left<#1\right>}}
  \newcommand{\lapla}{\nabla^{2}}

  \NewDocumentCommand{\relprime}{o}{%
    \IfNoValueTF{#1}{%
      \ensuremath{^{\prime}}%
    }{%
      \ensuremath{^{\prime{#1}}}%
    }%
  }

  \NewDocumentCommand{\ud}{o}{%
    \IfNoValueTF{#1}{%
      \ensuremath{\mathrm{d}}%
    }{%
```

```
    \ensuremath{\mathrm{d}^{#1}}%
  }%
}

% Total derivatives.
\NewDocumentCommand{\dd}{m m o}{%
  \IfNoValueTF{#3}{%
    \frac{\ud #1}{\ud #2}%
  }{%
    \frac{\ud^{#3}#1}{\ud {#2}^{#3}}%
  }%
}

% Open total derivative operator.
\NewDocumentCommand{\dda}{m m o}{%
  \IfNoValueTF{#3}{%
    \frac{\ud}{\ud #2}{#1}%
  }{%
    \frac{\ud^{#3}}{\ud {#2}^{#3}}{#1}%
  }%
}

% Partial derivatives.
\NewDocumentCommand{\ddp}{m m o}{%
  \IfNoValueTF{#3}{%
    \frac{\partial #1}{\partial #2}%
  }{%
    \frac{\partial^{#3}#1}{\partial {#2}^{#3}}%
  }%
}

% Open partial derivative operator.
\NewDocumentCommand{\ddpa}{m m o}{%
  \IfNoValueTF{#3}{%
    \frac{\partial}{\partial #2}{#1}%
  }{%
    \frac{\partial^{#3}}{\partial {#2}^{#3}}{#1}%
  }%
}

% Inner product.
\NewDocumentCommand{\innp}{m m}{\left<{#1},{#2}\right>}

% Chi-square helpers.
\newcommand{\chisq}{\ensuremath{\chi^{2}}}
\newcommand{\chisqred}{\ensuremath{\chi^{2}_{\text{red}}}}


% Roman numerals.
\newcommand{\rmnum}[1]{\romannumeral #1}
\newcommand{\Rmnum}[1]{\expandafter\@slowromancap\romannumeral #1@}

% Slanted parallel marker
\newcommand{\slparallel}{\mathchoice
  {\mathrel{/\mkern-5mu/}}% display
  {\mathrel{/\mkern-5mu/}}% text
  {\mathrel{/\mkern-4mu/}}% script
  {\mathrel{/\mkern-3mu/}}% scriptscript
```

```
  }

  % Cartesian, cylindrical, and spherical unit vectors.
  \newcommand{\vi}{\ensuremath{\hat{e}_{1}}}
  \newcommand{\vj}{\ensuremath{\hat{e}_{2}}}
  \newcommand{\vk}{\ensuremath{\hat{e}_{3}}}
  \newcommand{\vs}{\ensuremath{\hat{e}_{s}}}
  \newcommand{\vp}{\ensuremath{\hat{e}_{p}}}
  \newcommand{\vperp}{\ensuremath{\hat{e}_{\perp}}}
  \newcommand{\vpar}{\ensuremath{\hat{e}_{\slparallel}}}
  \newcommand{\vphi}{\ensuremath{\hat{e}_{\phi}}}
  \newcommand{\vz}{\ensuremath{\hat{e}_{z}}}
  \newcommand{\vsl}{\ensuremath{\hat{e}_{s\relprime}}}
  \newcommand{\vphil}{\ensuremath{\hat{e}_{\phi\relprime}}}
  \newcommand{\vzl}{\ensuremath{\hat{e}_{z\relprime}}}
  \newcommand{\vr}{\ensuremath{\hat{e}_{r}}}
  \newcommand{\vtheta}{\ensuremath{\hat{e}_{\theta}}}

\fi

% Number equations, tables, and figures within sections.
\numberwithin{equation}{section}
\numberwithin{table}{section}
\numberwithin{figure}{section}
\renewcommand{\arraystretch}{1.5}

% ------------------------------------------------------------------------------
% Figures and floats
% ------------------------------------------------------------------------------

\ifrelatese@figures
  \RequirePackage{float}
  \RequirePackage{subcaption}
  \RequirePackage{placeins}
  \RequirePackage{pdflscape}
  \RequirePackage{svg}

  \captionsetup[subfigure]{font=footnotesize,format=hang}
  \captionsetup[subtable]{font=footnotesize,format=hang,position=above,singlelinecheck=false}

  % Table captions above tables. This reproduces the original float setup.
  \floatstyle{plaintop}
  \restylefloat{table}
\fi

% ------------------------------------------------------------------------------
% Tables
% ------------------------------------------------------------------------------

\ifrelatese@tables
  \RequirePackage{colortbl}
  \RequirePackage{makecell}
\fi

% ------------------------------------------------------------------------------
% Scientific writing: units and chemistry
% ------------------------------------------------------------------------------

```

```
\ifrelatese@siunitx
  \RequirePackage[detect-all]{siunitx}

  % Language-neutral defaults.
  \sisetup{%
    group-digits=none,
    range-phrase=--,
    per-mode=symbol
  }

  % Decimal comma convention.
  \ifrelatese@sciencecomma
    \sisetup{output-decimal-marker={,}}
  \fi
\fi

\ifrelatese@chemistry
  \RequirePackage[version=4]{mhchem}
\fi

% ------------------------------------------------------------------------------
% Hyperlinks, bibliography, and clever references
% ------------------------------------------------------------------------------

\ifrelatese@biblatex
  \RequirePackage{csquotes}
  \RequirePackage[
    backend=biber,
    style=numeric-comp,
    sorting=none,
    backref=true,
    autocite=superscript
  ]{biblatex}
\fi

% hyperref should be loaded late. cleveref should be loaded after hyperref.
\ifrelatese@refs
  \RequirePackage[
    unicode,
    bookmarks=true,
    breaklinks
  ]{hyperref}

  \hypersetup{%
    colorlinks=true,
    linkcolor=darkblue,
    urlcolor=darkblue,
    citecolor=darkblue
  }
\fi

\ifrelatese@cleveref
  \ifrelatese@refs\else
    \RequirePackage[
      unicode,
      bookmarks=true,
      breaklinks
    ]{hyperref}
```

```
    \hypersetup{%
      colorlinks=true,
      linkcolor=darkblue,
      urlcolor=darkblue,
      citecolor=darkblue
    }
  \fi

  \RequirePackage{cleveref}
\fi

% ---------------------------------------------------------------------------
% Draft tools and small utilities
% ---------------------------------------------------------------------------

\ifrelatese@drafttools
  \RequirePackage{lipsum}
\fi

% Quotation helper.
\newcommand{\quoteblock}[2]{%
  \begin{samepage}%
    \begin{quotation}%
      \emph{``#1''}%
      \begin{flushright}%
        -- #2%
      \end{flushright}%
    \end{quotation}%
  \end{samepage}%
}

% Easy-to-find revision marker preserved from the original package.
\newcommand{\coxa}{%
  {\Huge C\kern-.1667em\lower.5ex\hbox{O}\-X\kern-.1667em\lower.5ex\hbox{A}\@}%
  \@ifundefined{index}{}{\index{CoXa}}%
}

\makeatother
\endinput
```